

mb-pc40s-2026-jan

Typst conversion review. OCR content and mathematical accuracy require editorial review.

Question 1

Liam went for a ride on a Ferris wheel with a radius of 5 metres. Determine the distance he travelled if the Ferris wheel stopped after rotating 1260° .

Solution

$$\theta = 1260^\circ \cdot \frac{\pi}{180^\circ} \quad 1 \text{ mark for conversion}$$

$$\theta = 7\pi$$

or

$$\theta = 21.991148\dots$$

$$s = \theta r$$

$$s = 7\pi(5)$$

$$s = 35\pi \text{ metres}$$

or

$$s = 109.956 \text{ metres}$$

1 mark for substitution

2 marks

Question 2

Determine and simplify the term that contains x^{12} in the binomial expansion of $(2x^4 - 3)^7$.

Solution

Method 1

$$\begin{aligned} & (x^4)^7, (x^4)^6, (x^4)^5, \dots \\ & x^{28}, x^{24}, x^{20}, \dots \\ t_5 &= {}_7C_4 (2x^4)^3 (-3)^4 \\ &= 22680x^{12} \end{aligned}$$

1 mark for determining a pattern

2 marks (1 mark for ${}_7C_4$; $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark
for exponent)

4 marks

Method 2

$$\begin{aligned} x^{12} &= (x^4)^{7-k} (x^0)^k \\ 12 &= 28 - 4k \\ -16 &= -4k \\ 4 &= k \\ t_5 &= {}_7C_4 (2x^4)^3 (-3)^4 \\ &= 22680x^{12} \end{aligned}$$

$\frac{1}{2}$ mark for substitution

$\frac{1}{2}$ mark for solving for k

2 marks (1 mark for ${}_7C_4$; $\frac{1}{2}$ mark for each consistent factor)

1 mark for simplification ($\frac{1}{2}$ mark for coefficient, $\frac{1}{2}$ mark
for exponent)

4 marks

Question 3

The normal healing of wounds can be modeled by an exponential function. The area of the wound decreases according to the formula,

$$A = A_0 e^{-0.35t}$$

where: A is the area of the wound in cm^2 , after t days,

A_0 is the initial area of the wound in cm^2 , and

t is the time, in days, after healing begins.

If a wound has an initial area of 25 cm^2 , determine:

a) the area of the wound after 3 days.

b) the amount of time it will take for the area of the wound to decrease to 4 cm^2 .

Solution

a)

$$A = 25e^{-0.35(3)}$$

$$A = 8.748 \text{ cm}^2$$

1 mark

b)

$$4 = 25e^{-0.35t}$$

$$0.16 = e^{-0.35t}$$

$$\ln 0.16 = \ln e^{-0.35t}$$

$$\ln 0.16 = -0.35t \ln e$$

$$\frac{\ln 0.16}{-0.35} = t$$

$$t = 5.235947...$$

$$t = 5.236 \text{ days} \quad 1/2 \text{ mark for the value of } t$$

1 mark for applying logarithms

1/2 mark for power law

2 marks

Question 4

During a hockey practice, 4 players are wearing a green jersey and 5 players are wearing a red jersey. Determine the number of ways three players can be selected if at least two of the players must be wearing a red jersey.

Solution

Case 1:

$${}_5C_2 \cdot {}_4C_1 = 40$$

1 mark for Case 1

Case 2:

$${}_5C_3 \cdot {}_4C_0 = 10$$

1 mark for Case 2

$$40 + 10 = 50 \text{ ways}$$

1 mark for addition of cases

3 marks

Note:

${}_4C_0$ does not need to be shown.

Question 5

Solve, algebraically, over the interval $[0, 2\pi]$.

$$2 \sin^2 x + 5 \cos x + 1 = 0$$

Solution

$$2(1 - \cos^2 x) + 5 \cos x + 1 = 0$$

$$2 - 2 \cos^2 x + 5 \cos x + 1 = 0$$

$$-2 \cos^2 x + 5 \cos x + 3 = 0$$

$$2 \cos^2 x - 5 \cos x - 3 = 0$$

$$(2 \cos x + 1)(\cos x - 3) = 0$$

$$\cos x = -\frac{1}{2} \quad \cos x = 3$$

$$x = \frac{2\pi}{3}, \frac{4\pi}{3} \quad \text{no solution}$$

1 mark for substitution of an appropriate identity

1 mark for solving for $\cos x$

2 marks for solving for x ($\frac{1}{2}$ mark for each value; 1 mark

for indicating no solution)

4 marks

Question 6

The function, $y = f(x)$, has a domain of $[-8, 4]$. The domain of the graph of $y = f(bx)$ is $[-4, 2]$.

State the value of b .

Solution

$$b = 2$$

1 mark

Question 7

Describe the difference between the graph of $f(x) = \frac{(x+3)(x-4)}{(x+3)}$ and the graph of $g(x) = x - 4$.

Solution

The graph of $f(x)$ has a point of discontinuity at $x = -3$, where as the graph of $g(x)$ does not.

1 mark

Question 8

Determine if the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is on the unit circle. Justify your answer.

Solution

Method 1

$$x^2 + y^2 = 1$$

$$\begin{aligned}\text{Left-Hand Side} &= \left(-\frac{3}{5}\right)^2 + \left(\frac{2}{\sqrt{3}}\right)^2 \\ &= \frac{9}{25} + \frac{4}{3} \\ &= \frac{27}{75} + \frac{100}{75}\end{aligned}$$

$$\frac{127}{75} \neq 1$$

Therefore the point $\left(-\frac{3}{5}, \frac{2}{\sqrt{3}}\right)$ is not on the unit circle.

1 mark for justification

1 mark

Method 2

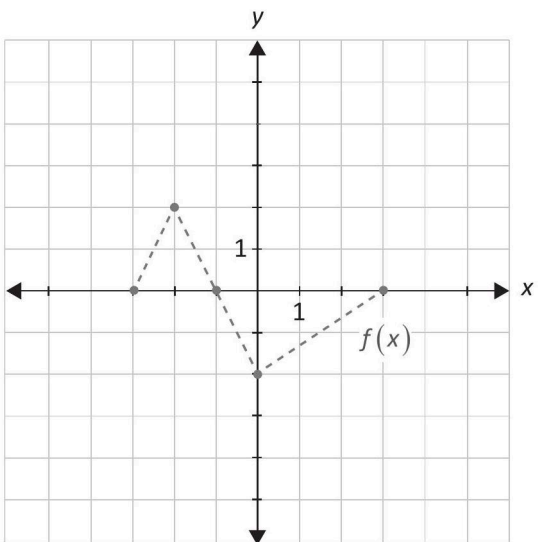
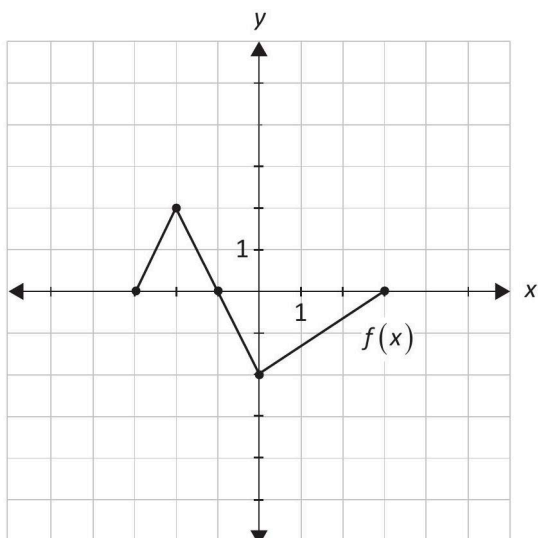
Since $\frac{2}{\sqrt{3}} > 1$, the point is not on the unit circle.

1 mark for justification

1 mark

Question 9

Given the graph of $f(x)$, sketch the graph of $g(x) = f(-(x - 1)) + 3$.



The graph of $f(x)$

has already been

drawn for your

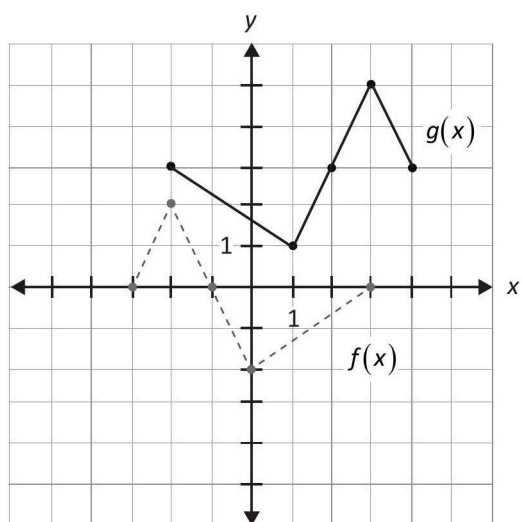
reference.

No marks will be

awarded for the

graph of $f(x)$.

Solution



1 mark for horizontal reflection

1 mark for horizontal translation

1 mark for vertical translation

3 marks

Question 10

Bonnie was asked to determine the factors of $p(x) = -3x^4 + 2x^2 - 4x + 5$.

Describe the error she made in her set up of synthetic division.

$1 \overline{) -3 \ 2 \ -4 \ 5}$

Solution

Bonnie did not write the coefficient of 0 for the missing term, x^3 .

1 mark

Question 11

Given the functions, $f(x) = \frac{1}{x-1}$ and $g(x) = x - 3$,

a) state the equation of $f(g(x))$.

$$f(g(x)) =$$

b) state the domain of $f(g(x))$.

Domain:

Solution

a) $f(g(x)) = \frac{1}{x-4}$

1 mark

b) Domain:

$$\{x \mid x \neq 4, x \in \mathbb{R}\}$$

or

Domain:

Question 12

Prove the identity for all permissible values of x .

$$\tan 2x = \frac{2}{\cot x - \tan x}$$

Left-Hand Side	Right-Hand Side

Solution

Method 1

Left-Hand Side	Right-Hand Side
$\tan 2x$	$\frac{2}{\frac{\cos x}{\sin x} - \frac{\sin x}{\cos x}}$
	$\frac{2}{\frac{\cos^2 x - \sin^2 x}{\sin x \cos x}}$
	$\frac{2 \sin x \cos x}{\cos^2 x - \sin^2 x}$
	$\frac{\sin 2x}{\cos 2x}$
	$\tan 2x$

1 mark for correct substitution of appropriate identities

1 mark for algebraic strategies

1 mark for logical process to prove the identity

3 marks

Question 13

Given $f(x) = -x^2 + 2$ and $g(x) = x^2 - 1$,

a) determine the equation of $h(x) = g(x) - f(x)$.

$h(x) =$

b) state the range of $h(x)$.

Range:

Solution

a) $h(x) = g(x) - f(x)$

$$h(x) = (x^2 - 1) - (-x^2 + 2)$$

$$h(x) = x^2 - 1 + x^2 - 2$$

$$h(x) = 2x^2 - 3$$

b) Range:

$$\{y \mid y \geq -3, y \in \mathbb{R}\}$$

or

Range:

$$[-3, \infty)$$

1 mark for subtraction of $g(x) - f(x)$

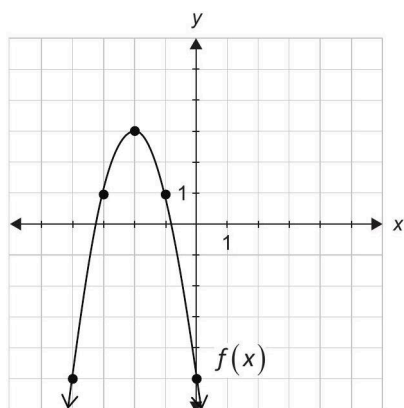
1 mark

1 mark for range consistent with $h(x)$

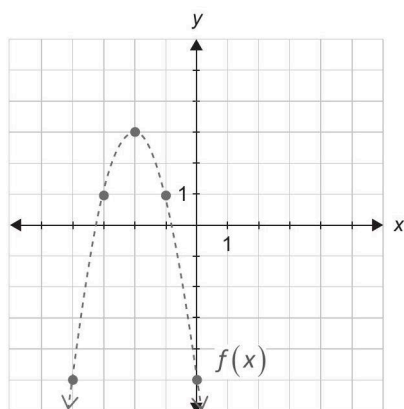
1 mark

Question 14

Given the graph of $f(x)$,



a) sketch the graph after a reflection over the line $y = x$.



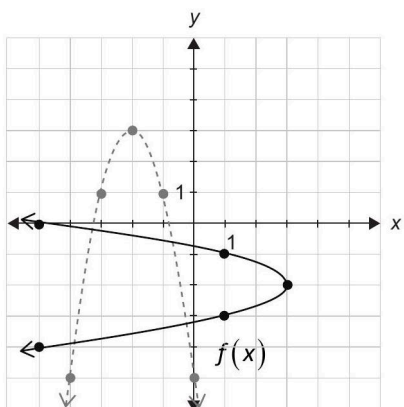
The graph of $f(x)$
has already been
drawn for your
reference.

No marks will be
awarded for the
graph of $f(x)$.

b) state a restriction on the domain of $f(x)$ in order for its inverse to be a function.

Solution

a)



1 mark

b) $\{x \mid x \geq -2, x \in \mathbb{R}\}$

or

$$\{x \mid x \leq -2, x \in \mathbb{R}\}$$

1 mark

Note:

Other answers are possible for b).

Question 15

Describe a transformation of a function that would not affect its domain.

Solution

vertical stretch by a factor of a

or

vertical reflection

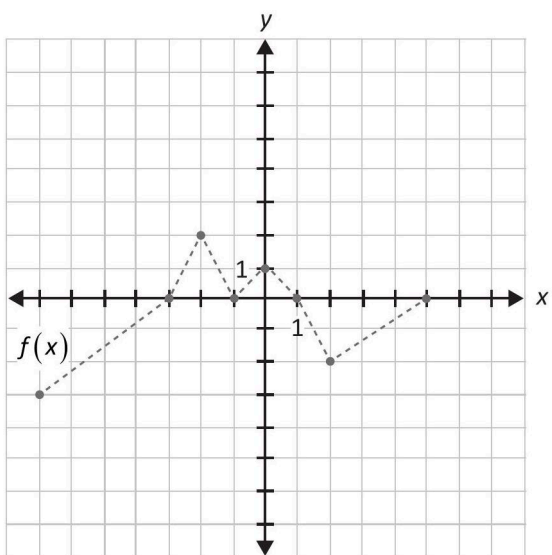
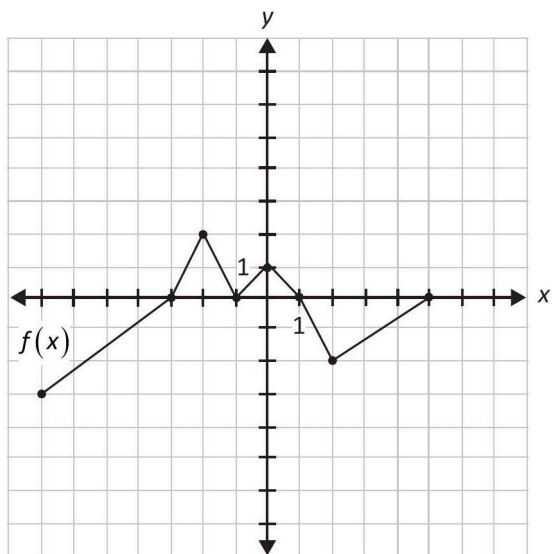
or

vertical translation k units up/down

1 mark

Question 16

Given the graph of $f(x)$, sketch the graph of the function $y = |f(x)|$.



The graph of $f(x)$

has already been

drawn for your

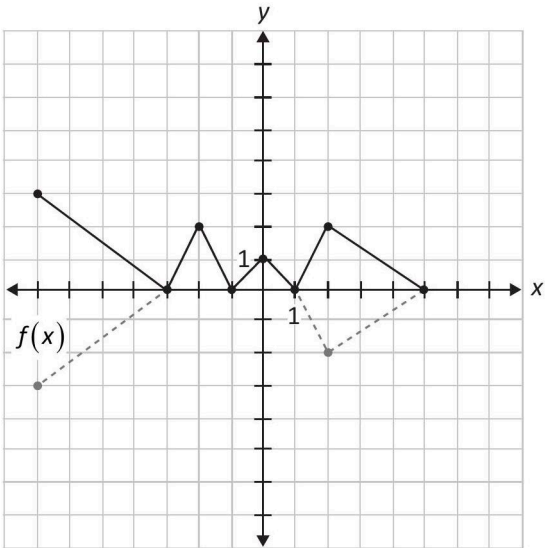
reference.

No marks will be

awarded for the

graph of $f(x)$.

Solution



1 mark

Question 17

Determine, algebraically, all of the zeros of the polynomial function, $p(x) = 3x^3 + x^2 - 20x + 12$.

Solution

$$p(x) = 3x^3 + x^2 - 20x + 12$$

$$p(2) = 3(2)^3 + (2)^2 - 20(2) + 12$$

$$p(2) = 0$$

$$\therefore (x - 2) \text{ is a factor}$$

2	3	1	-20	12
		6	14	-12
	3	7	-6	0

1 mark for synthetic division (or equivalent strategy)

$$p(x) = (x - 2)(3x^2 + 7x - 6)$$

$$0 = (x - 2)(3x - 2)(x + 3)$$

$$x = 2, \quad x = \frac{2}{3}, \quad x = -3$$

$\frac{1}{2}$ mark for consistent factors

$\frac{1}{2}$ mark for consistent zeros

3 marks

Question 18

Given $\theta = 240^\circ$, identify the coordinates of the point, $P(\theta)$, on the unit circle.

A.

$$\left(-\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$$

B.

$$\left(-\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$$

C.

$$\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

D.

$$\left(\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

Solution

Correct answer: C

$$\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$$

Question 19

Given the functions, $f(x) = -3x + 5$ and $g(x) = x^2 + x - 1$, identify the value of $g(f(2))$.

A.

-10

B.

-5

C.

-3

D.

-1

Solution

Correct answer: D

-1

Question 20

Given $f(x) = \sqrt{x} + 5$, identify the equation of the transformed graph that has the same y -intercept.

A.

$$y = f(x - 3)$$

B.

$$y = -3f(x)$$

C.

$$y = f(-3x)$$

D.

$$y = f(x) - 3$$

Solution

Correct answer: C

$$y = f(-3x)$$

Question 21

Given the graph of $p(x) = ax^3 + bx^2 + cx + d$, identify the statement that is true.

A.

$$a > 0, d < 0$$

B.

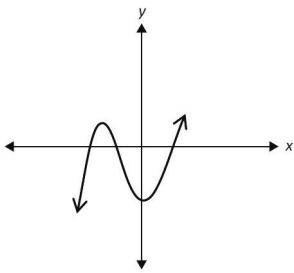
$$a > 0, d > 0$$

C.

$$a < 0, d < 0$$

D.

$$a < 0, d > 0$$



Solution

Correct answer: A

$$a > 0, d < 0$$

Question 22

Identify the general solution of the equation, $\csc \theta = -1$.

A.

$$\theta = \frac{\pi}{2} + k\pi, k \in \mathbb{Z}$$

B.

$$\theta = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

C.

$$\theta = \frac{3\pi}{2} + k\pi, k \in \mathbb{Z}$$

D.

$$\theta = \pi + 2k\pi, k \in \mathbb{Z}$$

Solution

Correct answer: B

$$\theta = \frac{3\pi}{2} + 2k\pi, k \in \mathbb{Z}$$

Question 23

Identify the equation of a radical function with a domain of $[-6, \infty)$ and a range of $(-\infty, 3]$.

A.

$$y = -\sqrt{x+6} - 3$$

B.

$$y = -\sqrt{x+6} + 3$$

C.

$$y = \sqrt{-(x+3)} - 6$$

D.

$$y = \sqrt{x-6} + 3$$

Solution

Correct answer: B

$$y = -\sqrt{x+6} + 3$$

Question 24

1 mark

Identify the value of n in the equation, ${}_nC_3 = {}_nC_7$.

A.

3

B.

4

C.

7

D.

10

Solution

Correct answer: D

10

Question 25

Identify the remainder when $p(x) = x^5 - 1$ is divided by $(x + 1)$.

A.

-2

B.

-1

C.

0

D.

4

Solution

Correct answer: A

-2

Question 26

Match each function with its corresponding graph.

Write the corresponding letter in this column.

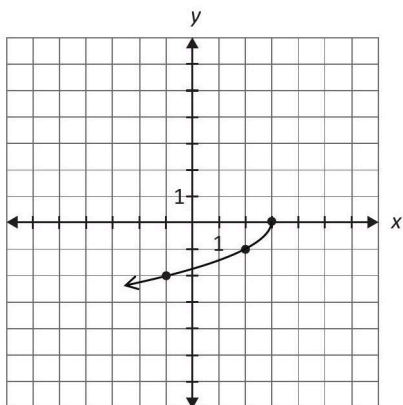
$$y = \sqrt{x} - 3$$

$$y = \sqrt{-x} + 3$$

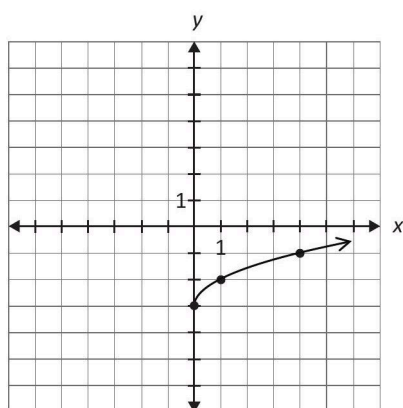
$$y = -\sqrt{x + 3}$$

$$y = -\sqrt{-(x - 3)}$$

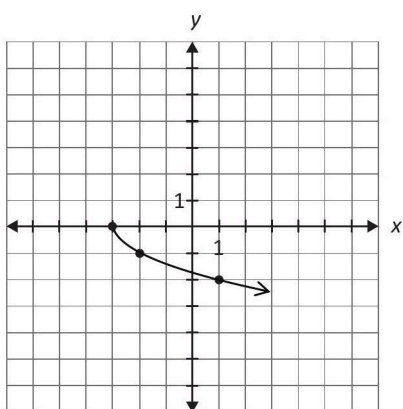
A)



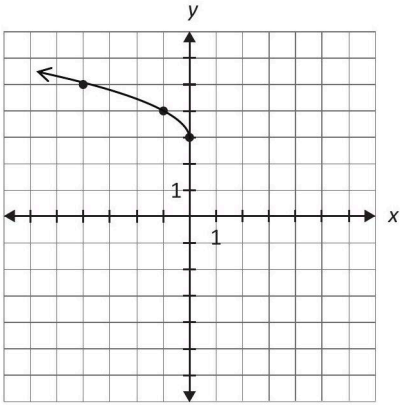
B)



C)



D)



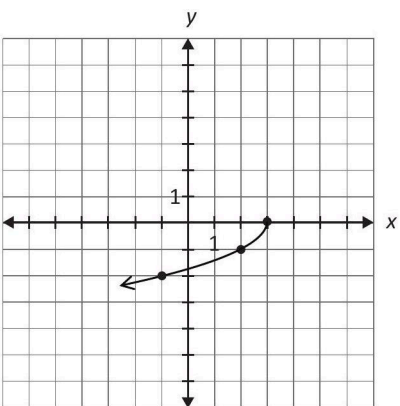
Solution

C

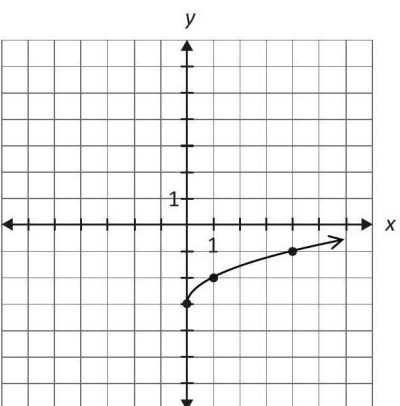
$\frac{1}{2}$ mark for each correct answer

2 marks

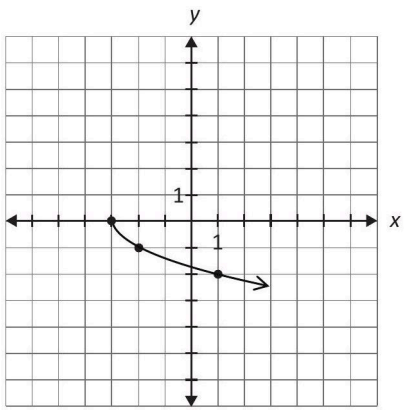
A)



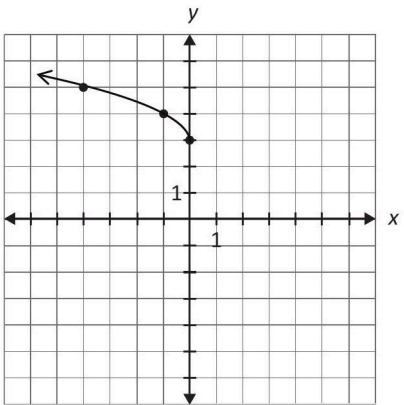
B)



C)



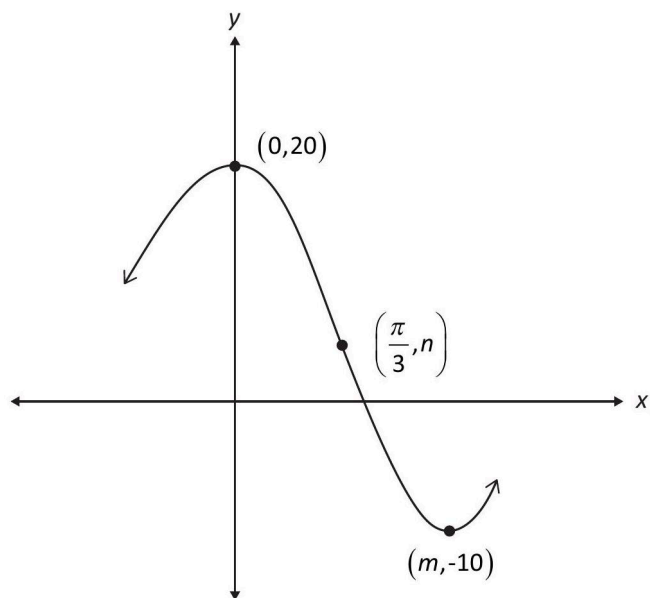
D)



Question 27

The amplitude, A , of the sinusoidal function can be determined using the equation $A = 20 - n$.

State the values of m and n .



Solution

$$m = \frac{2\pi}{3}$$

$$n = 5$$

1 mark for value of m

1 mark for value of n

2 marks

Question 28

Solve, algebraically.

$${}_{n-1}P_2 = 20$$

Solution

$$\frac{(n-1)!}{(n-1-2)!} = 20$$

$$\frac{(n-1)!}{(n-3)!} = 20$$

$$\frac{(n-1)(n-2)(n-3)!}{(n-3)!} = 20$$

$$n^2 - 3n + 2 = 20$$

$$n^2 - 3n - 18 = 0$$

$$(n-6)(n+3) = 0$$

$$n = 6 \quad n \geq -3$$

$\frac{1}{2}$ mark for substitution

1 mark for factorial expansion

$\frac{1}{2}$ mark for simplification of factorials

$\frac{1}{2}$ mark for the permissible value of n

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

3 marks

Question 29

Determine the exact value of $\sin\left(\frac{23\pi}{12}\right)$.

Solution

$$\begin{aligned}\sin\left(\frac{23\pi}{12}\right) &= \sin\left(\frac{21\pi}{12} + \frac{2\pi}{12}\right) \\ \sin\left(\frac{7\pi}{4} + \frac{\pi}{6}\right) &= \sin\left(\frac{7\pi}{4}\right) \cos\left(\frac{\pi}{6}\right) + \cos\left(\frac{7\pi}{4}\right) \sin\left(\frac{\pi}{6}\right) \\ &= \left(-\frac{\sqrt{2}}{2}\right) \left(\frac{\sqrt{3}}{2}\right) + \left(\frac{\sqrt{2}}{2}\right) \left(\frac{1}{2}\right) \\ &= \frac{-\sqrt{6} + \sqrt{2}}{4}\end{aligned}$$

1 mark for substitution into correct identity

2 marks for exact values ($\frac{1}{2}$ mark for each)

3 marks

Note:

Other combinations are possible.

Question 30

Cameron was asked to solve the equation, $\log_3(x - 4) = 2$.

Cameron's solution:

$$x - 4 = 2^3$$

$$x - 4 = 8$$

$$x = 12$$

Describe Cameron's error.

Solution

Cameron mixed up the exponent and the base when changing to exponential form.

1 mark

Question 31

Given the identity $\sin^2 \theta = \frac{\tan \theta \sin \theta}{\sec \theta}$, state the non-permissible value of θ over the interval $[\pi, 2\pi]$.

Solution

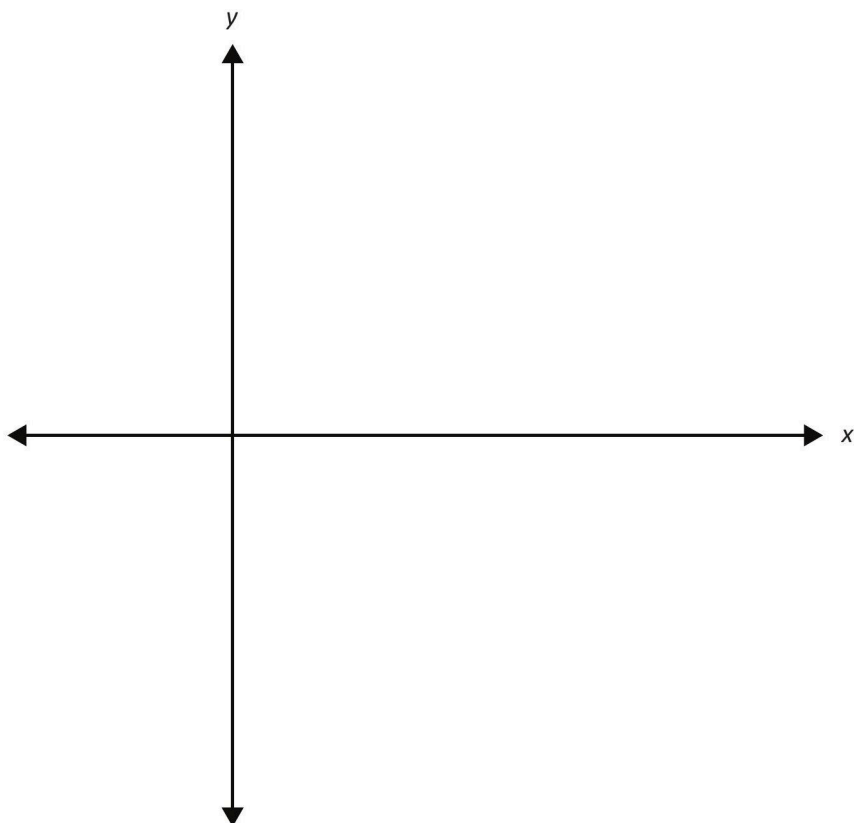
$$\cos \theta = 0$$

$$\theta = \frac{3\pi}{2}$$

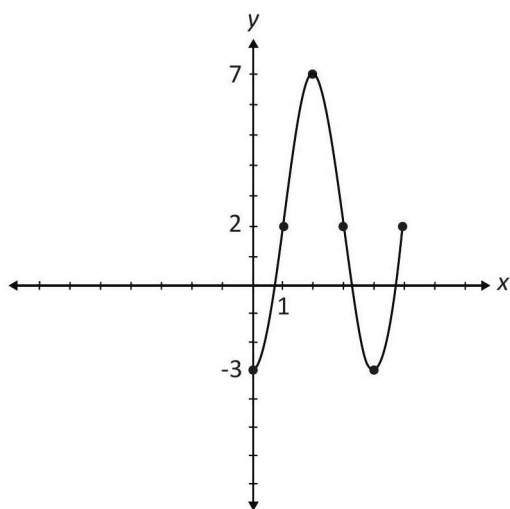
1 mark

Question 32

Sketch the graph of $y = -5 \cos\left(\frac{\pi}{2}x\right) + 2$ over the domain $[0, 5]$.



Solution



1 mark for shape of a sinusoidal function with correct amplitude

1 mark for vertical reflection

1 mark for vertical translation

1 mark for period

4 marks

Question 33

Describe the behaviour of the graph of the polynomial function, $p(x) = (x - 3)^2(x - 1)$, as it approaches the x -intercept at $x = 3$.

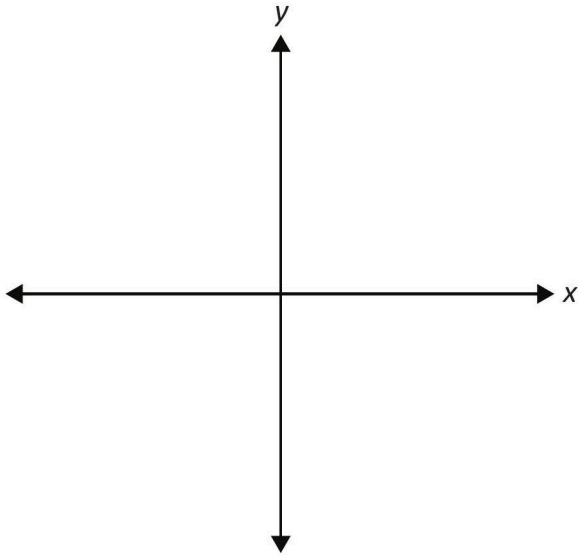
Solution

As it approaches $x = 3$, the graph touches but does not cross the x -axis.

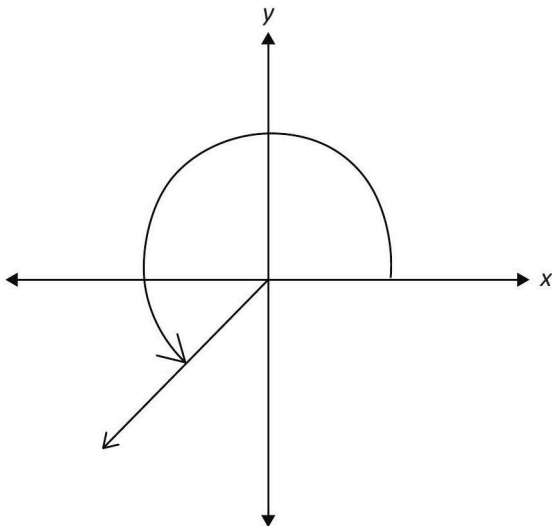
1 mark

Question 34

Sketch the angle of 4 radians in standard position.



Solution



$\frac{1}{2}$ mark for an appropriate angle

in quadrant III

$\frac{1}{2}$ mark for the correct direction of an

angle in standard position

1 mark

Question 35

Solve, algebraically.

$$\log_5 12 + 2 \log_5 x = \log_5 48$$

Solution

Method 1

$$\begin{aligned}\log_5 12 + 2 \log_5 x &= \log_5 48 \\ \log_5 12 + \log_5 x^2 &= \log_5 48 \\ \log_5 12x^2 &= \log_5 48 \\ 12x^2 &= 48 \\ x^2 &= 4 \\ x = 2 &\quad x < -2\end{aligned}$$

$\frac{1}{2}$ mark for permissible value of x

$\frac{1}{2}$ mark for showing the rejection of the extraneous root

4 marks

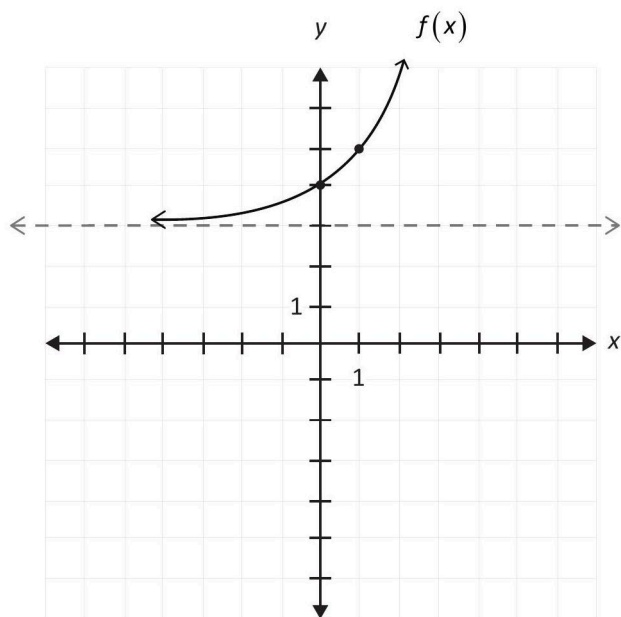
Method 2

$$\begin{aligned}\log_5 12 + 2 \log_5 x &= \log_5 48 \\ \log_5 12 + \log_5 x^2 - \log_5 48 &= 0 && 1 \text{ mark for power law} \\ \log_5 \left(\frac{12x^2}{48} \right) &= 0 && \begin{array}{l} 1/2 \text{ mark for product law} \\ 1/2 \text{ mark for quotient law} \end{array} \\ \frac{12x^2}{48} &= 5^0 && 1 \text{ mark for exponential form} \\ x^2 &= 4\end{aligned}$$

4 marks

Question 36

State the equation of the exponential function represented by the graph of $f(x)$.



$f(x) =$

—

Solution

$$f(x) = 2^x + 3$$

1 mark for exponential function with a base of 2

1 mark for vertical translation

2 marks

Question 37

Evaluate.

$$\cos\left(\frac{11\pi}{3}\right) + \csc\left(-\frac{\pi}{3}\right) \cot\left(\frac{11\pi}{6}\right)$$

Solution

$$\left(\frac{1}{2}\right) + \left(-\frac{2}{\sqrt{3}}\right)\left(-\frac{\sqrt{3}}{1}\right)$$

$$\frac{1}{2} + 2$$

$$\frac{5}{2}$$

1 mark for $\cos\left(\frac{11\pi}{3}\right)$ (1/2 mark for value; $\frac{1}{2}$ mark for quadrant)

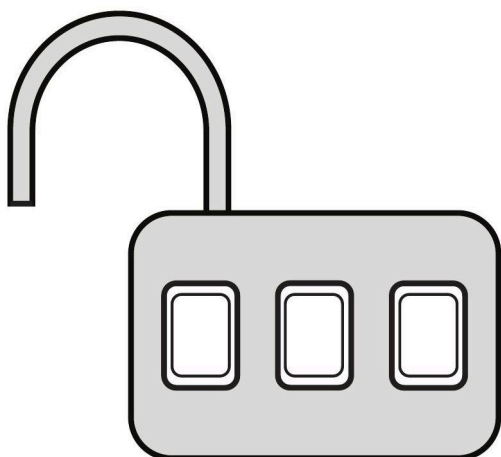
1 mark for $\csc\left(-\frac{\pi}{3}\right)$ ($\frac{1}{2}$ mark for value; $\frac{1}{2}$ mark for quadrant)

1 mark for $\cot\left(\frac{11\pi}{6}\right)$ (1/2 mark for value; $\frac{1}{2}$ mark for quadrant)

3 marks

Question 38

Min Li has a 3-digit code for his lock. Determine the number of possible codes for his lock, if repetition is allowed.



Solution

$$10 \cdot \underline{10} \cdot \underline{10} = 1000$$

1 mark

Question 39

Solve, algebraically.

$$\left(\frac{1}{6}\right)^{3x+2} = 6^{2x}$$

Solution

$$(6^{-1})^{3x+2} = 6^{2x}$$

$$6^{-3x-2} = 6^{2x}$$

$$-3x - 2 = 2x$$

$$-5x = 2$$

$$x = -\frac{2}{5}$$

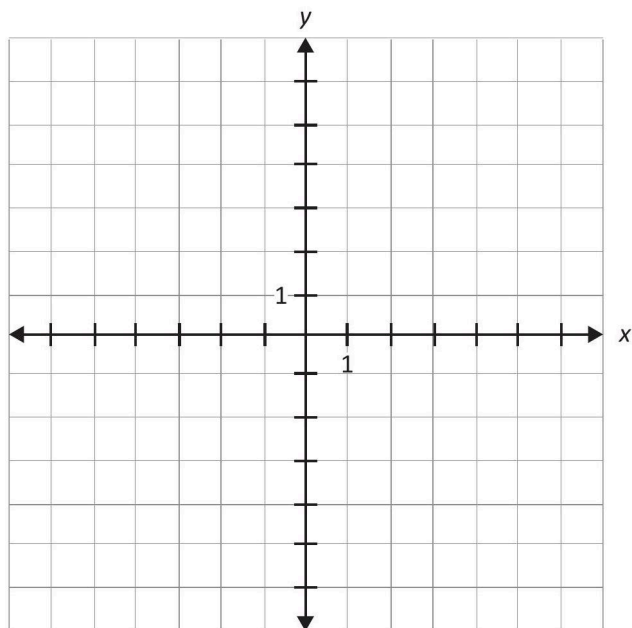
1 mark for changing to a common base

1 mark for equating exponents

2 marks

Question 40

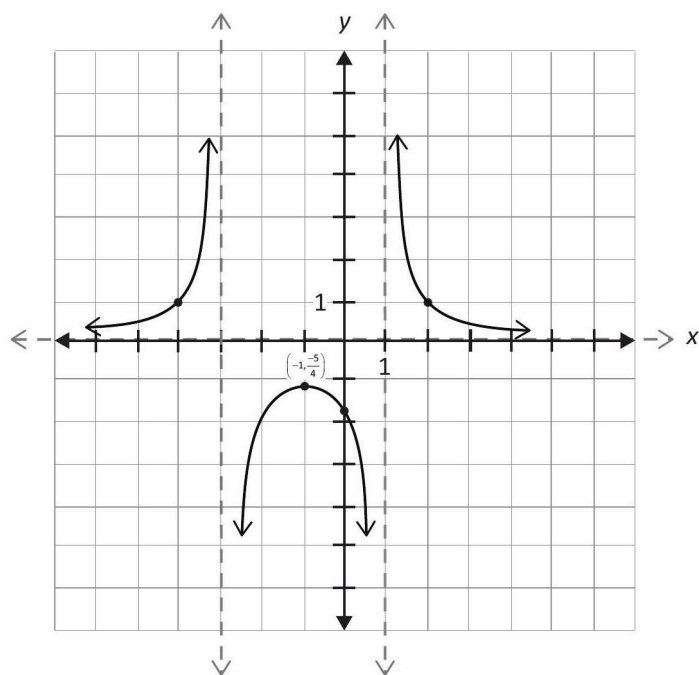
Sketch the graph of $f(x) = \frac{5}{(x-1)(x+3)}$ and state the y -intercept.



y -intercept:

—

Solution



$(0, -\frac{5}{3})$

—

1 mark for vertical asymptotic behaviour

($\frac{1}{2}$ mark for behaviour approaching $x = -3$; $\frac{1}{2}$ mark for behaviour approaching $x = 1$)

1 mark for horizontal asymptotic behaviour approaching $y = 0$

$1\frac{1}{2}$ marks for shape ($\frac{1}{2}$ mark for shape of each branch)

$\frac{1}{2}$ mark for y -intercept

4 marks

Question 41

Given $\log_2 x = 5$, determine the value of $\log_4(2x)$.

Solution

$$\begin{aligned}x &= 2^5 & 1 \text{ mark for solving for } x \\x &= 32\end{aligned}$$

$$\log_4(2(32))$$

$$\log_4(64)$$

3

1 mark for evaluating the logarithm

2 marks

Question 42

Henry was asked to prove the identity, $\csc^2 \theta - 2 \cos^2 \theta + 1 = \cot^2 \theta + 2 \sin^2 \theta$, for all permissible values of θ .

His work:

	Left-Hand side
$\csc^2\left(\frac{\pi}{4}\right) - 2 \cos^2\left(\frac{\pi}{4}\right) + 1$	$\cot^2\left(\frac{\pi}{4}\right) + 2 \sin^2\left(\frac{\pi}{4}\right)$
$(\sqrt{2})^2 - 2\left(\frac{\sqrt{2}}{2}\right)^2 + 1$	$(1)^2 + 2\left(\frac{\sqrt{2}}{2}\right)^2$
$2 - 1 + 1$	$1 + 1$
2	2
LHS	RHS ✓

Explain why his proof is insufficient.

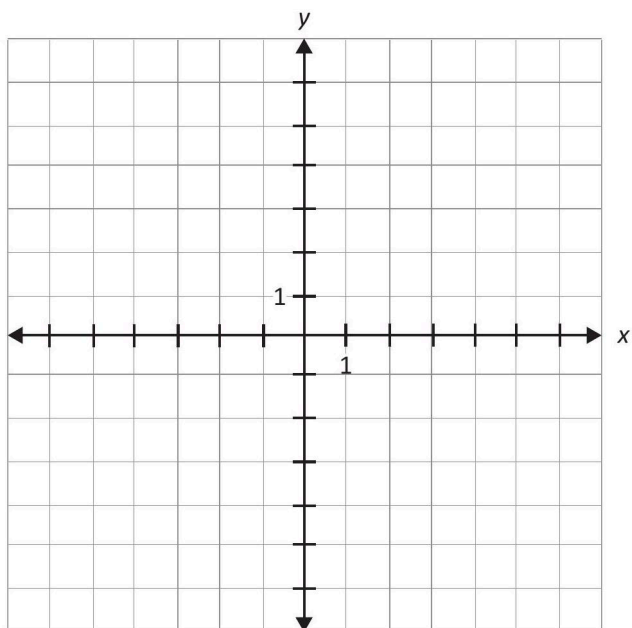
Solution

Henry proved the identity for $\theta = \frac{\pi}{4}$, but not for all permissible values of θ .

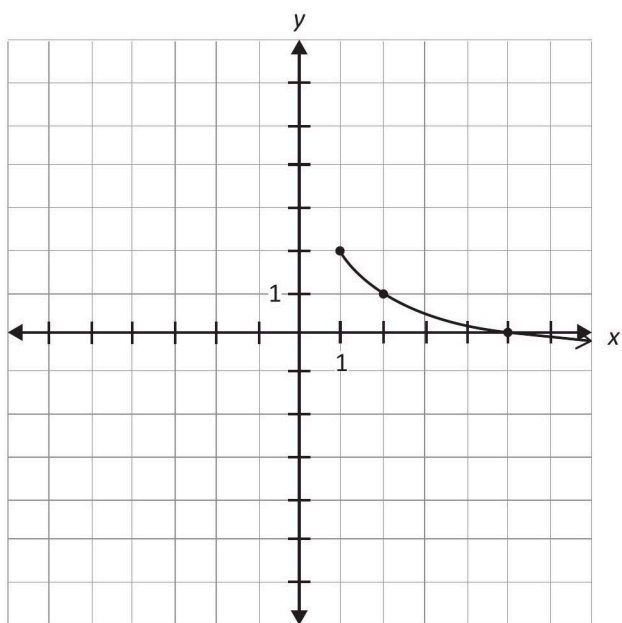
1 mark

Question 43

Sketch the graph of $y - 2 = -\sqrt{x - 1}$.



Solution



1 mark for shape of a radical function

1 mark for vertical reflection

1 mark for horizontal translation

1 mark for vertical translation

4 marks

Question 44

Evaluate.

$$\log_3 54 - \log_3 6$$

Solution

$$\log_3 \left(\frac{54}{6} \right)$$

$$\log_3 9$$

$$2$$

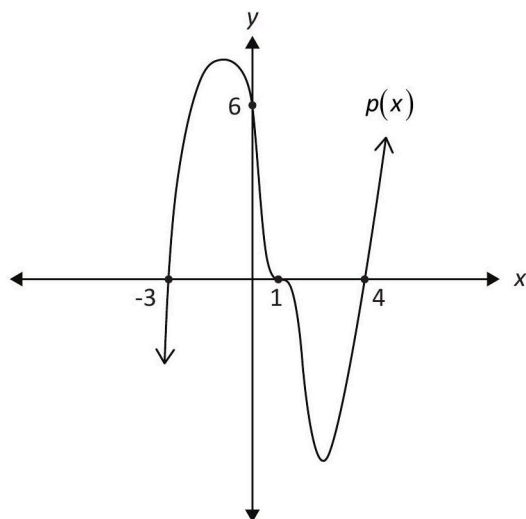
1 mark for quotient law

1 mark for evaluating the logarithm

2 marks

Question 45

Determine, algebraically, the equation of the graph of the polynomial function, $p(x)$.



$$p(x) =$$

Solution

$$p(x) = a(x - 1)^3(x + 3)(x - 4)$$

$$p(0) = a(0 - 1)^3(0 + 3)(0 - 4)$$

$$6 = a(-1)^3(3)(-4)$$

$$6 = 12a$$

$$\frac{1}{2} = a$$

$$p(x) = \frac{1}{2}(x - 1)^3(x + 3)(x - 4)$$

$\frac{1}{2}$ mark for factors of $p(x)$

$\frac{1}{2}$ mark for multiplicity of 3 at $x = 1$

1 mark for consistent value of a

2 marks